CSE 400 Lecture 16 Scribe

L15_L17.pdf
PDF

Generate a lecture scribe intended to serve as exam-oriented reference material for CSE 400: Fundamentals of Probability in Computing, Lecture 16.

Use ONLY the provided context (Lecture 16 PPT). Do not use any CP questions or any material not explicitly present in the lecture slides or the relevant textbook sections covered in this lecture.

The scribe must strictly follow the lecture's structure, scope, notation, slide order, and sequence exactly as presented in the PPT. Prioritize correctness and faithfulness to the lecture content. Do not invent definitions, results, proofs, examples, explanations, intuition, or interpretations that are not explicitly present in the slides.

While writing the scribe, reason step by step through all definitions, assumptions, theorem statements, proofs, derivations, and worked examples exactly as shown in the lecture slides. Make every logical and algebraic step explicit if it appears in the slides, so the document can be reliably used for closed-notes exam revision.

Include:
- All slide titles and section headings in order
- All definitions and notation introduced
- All stated assumptions and conditions
- All theorems or results exactly as written
- Step-by-step proofs or derivations where they appear
- Worked examples with all intermediate steps, only if they are part of the lecture

Do not simplify, reinterpret, summarize, compress, or add explanations beyond what is explicitly covered in the slides. Do not introduce new examples or

alternative solution methods.

Organize the scribe clearly using structured section headings corresponding to the lecture slides so that a student studying only this document can accurately revise Lecture 16 for an exam.

Below is an **exam-oriented lecture scribe for CSE400 – Lecture 16** constructed strictly from the content visible in the provided slides. The structure, headings, notation, and ordering follow the lecture slides. All definitions, steps, and examples included are only those that appear in the slides. Citations reference the uploaded lecture file.

CSE 400 – Fundamentals of Probability in Computing

Lecture 16 Scribe: Joint Random Variables and Joint Distributions

Based strictly on the lecture slides by **Dhaval Patel, PhD** (SEAS – Ahmedabad University).

---

1. Why Study Joint Random Variables?

In science and real life, we are often interested in **two or more random variables simultaneously** rather than studying a single random variable in isolation.

Examples of pairs of variables studied jointly:

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members

Question posed in lecture:
What type of relationship would be expected between each pair of variables? Why?

---

2. Applications Motivating Joint Random Variables

Example 1: Smart Transportation

Two random variables:

- X = Vehicle Speed
- Y = Traffic Density

Question posed:

> Can speed be high when the traffic density is also high?

Answer (conceptually posed):
Yes / No

Conclusion:

- X and Y are **correlated random variables**.

Applications in Intelligent Transportation Systems (ITS):

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

Example 2: Wireless Network Performance

Two important random variables:

- X = Signal-to-Noise Ratio (SNR)
- Y = Data Throughput

Observation:

- Throughput depends on SNR.
- Both vary due to **fading, interference, and congestion**.

Question:

> If SNR is high, will throughput always be high?

Answer presented:

- Yes / No

Reasoning:

- Network congestion can influence throughput even when SNR is high.

Example probability table shown in slides:

SNR Level	High Throughput	Low Throughput
High	0.40	0.10
Medium	0.20	0.15
Low	0.05	0.10

Applications:

- 5G scheduling
- Adaptive modulation
- Network planning

Additional Application Examples

Example 3:

- X = Rainfall
- Y = Temperature
  Application: Weather forecasting

Example 4:

- X = Wind Speed
- Y = Delivery Time
  Application: Logistics and UAV systems

Example 5:
Rolling two dice:

- Random variables X and Y represent outcomes of each die.

Example questions:

- $X + Y = 7$
- $X > Y$

Such questions require the **joint probability mass function (PMF)**.

---

3. Discrete Case – Joint PMF

The lecture introduces the **joint probability mass function (PMF)** for discrete random variables.

For two discrete random variables X and Y, probabilities are assigned to **pairs of values** (x, y).

The representation commonly used in the slides is a **Joint PMF Table**.

---

4. Joint PMF Table Representation

A **table representation** lists all possible values of X and Y and their joint probabilities.

Structure:

- Rows correspond to values of one random variable (e.g., X).
- Columns correspond to values of the other random variable (e.g., Y).
- Each cell contains

$$p(x, y) = P(X = x, Y = y)$$

This table shows the **probability associated with each pair of outcomes**.

---

5. Example: Rolling Two Dice

The lecture uses the example of **rolling two dice** to illustrate joint distributions.

Define:

- X = value of first die
- Y = value of second die

Possible outcomes:

$$(X, Y) \in \{1, 2, 3, 4, 5, 6\} \times \{1, 2, 3, 4, 5, 6\}$$

Total possible ordered pairs:

$$36$$

Each outcome:

$$P(X = x, Y = y) = \frac{1}{36}$$

Example events considered in lecture context:

- $X + Y = 7$
- $X > Y$

These probabilities can be computed using the **joint PMF table**.

---

6. Continuous Case – Joint PDF

For continuous random variables X and Y, the lecture introduces the **joint probability density function (PDF)**.

Denoted as:

$$f(x, y)$$

This density describes the probability distribution of the pair (X, Y).

The joint PDF must satisfy:

1. Non-negativity

$$f(x, y) \geqslant 0$$

2. Normalization condition

$$\int \int f(x, y)\, dx\, dy = 1$$

The lecture also provides a **geometric interpretation** using areas in the $x - y$ plane.

---

7. Worked Example (Continuous Joint PDF)

Given:

$$f(x, y) = 4xy, \quad 0 < x < 1,\ 0 < y < 1$$

Thus, (X, Y) takes values in the **unit square**:

$$[0, 1] \times [0, 1]$$

Tasks:

1. Show that $f(x, y)$ is a valid joint PDF.
2. Visualize the event

$$A = \{X < 0.5, Y > 0.5\}$$

3. Compute $P(A)$.

---

8. Example Solution

Step 1: Check Validity (Normalization)

We verify:

$$\int_0^1 \int_0^1 4xy \, dx \, dy$$

First integrate with respect to x:

$$\int_0^1 4xy \, dx = 4y \int_0^1 x \, dx$$

$$= 4y \left[\frac{x^2}{2}\right]_0^1 = 4y \cdot \frac{1}{2} = 2y$$

Now integrate with respect to y:

$$\int_0^1 2y \, dy$$

$$= [y^2]_0^1 = 1$$

Thus:

$$\int_0^1 \int_0^1 4xy \, dx \, dy = 1$$

Conclusion:

- $f(x, y) \geqslant 0$
- Total probability = 1

Therefore, $f(x, y)$ **is a valid joint PDF**.

Step 2: Compute $P(A)$

Event:

$$A = \{X < 0.5, Y > 0.5\}$$

Probability:

$$P(A) = \int_0^{0.5} \int_{0.5}^{1} 4xy \, dy \, dx$$

First integrate with respect to y:

$$\int_{0.5}^{1} 4xy \, dy = 4x \int_{0.5}^{1} y \, dy$$

$$= 4x \left[\frac{y^2}{2}\right]_{0.5}^{1}$$

$$= 4x \left(\frac{1 - 0.25}{2}\right)$$

$$= 4x \left(\frac{0.75}{2}\right)$$

$$= 4x \cdot 0.375 = 1.5x$$

Now integrate with respect to x:

$$P(A) = \int_0^{0.5} 1.5x \, dx$$

$$= 1.5 \left[\frac{x^2}{2}\right]_0^{0.5}$$

$$= 1.5 \cdot \frac{0.25}{2}$$

$$= \frac{3}{16}$$

Thus:

$$P(A) = \frac{3}{16}$$

---

9. Joint Cumulative Distribution Function (Joint CDF)

Suppose X and Y are jointly distributed random variables.

Notation:

$$X < x, Y < y$$

means the event

$$\{X < x \text{ and } Y < y\}$$

Definition:

The **Joint Cumulative Distribution Function** is

$$F(x, y) = P(X < x, Y < y)$$

---

10. Joint CDF – Continuous Case

If X and Y are continuous random variables with joint density $f(x, y)$ over a region $[a, b] \times [c, d]$, then

$$F(x, y) = \int \int f(u, v)\, du\, dv$$

Thus, the joint CDF is obtained by **integrating the joint density**.

Recovering the joint PDF:

Since there are two variables, we use **partial derivatives**:

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

---

11. Joint CDF – Discrete Case

If X and Y are discrete random variables with joint PMF $p(x_i, y_j)$, the joint CDF is obtained by **summing probabilities**:

$$F(x, y) = P(X \leqslant x, Y \leqslant y)$$

which is computed by summing

$$p(x_i, y_j)$$

over all pairs satisfying

$$x_i \leqslant x,\ y_j \leqslant y$$

---

12. Properties of the Joint CDF

The lecture introduces properties of the joint CDF (as listed in the slides):

1. Defined as

$$F(x, y) = P(X < x, Y < y)$$

2. Values lie within

$$0 \leqslant F(x, y) \leqslant 1$$

3. The function is **non-decreasing** in each argument.
4. It approaches **1** as x and y approach their maximum values.
5. It approaches **0** as x and y approach their minimum values.

End of Lecture 16 Topics

Topics covered in this lecture:

1. Motivation for studying joint random variables
2. Joint PMF (discrete case)
3. Table representation of joint distributions
4. Example using two dice
5. Joint PDF (continuous case)
6. Worked example with density $f(x, y) = 4xy$
7. Joint cumulative distribution function (Joint CDF)
8. Continuous and discrete formulations of the Joint CDF
9. Properties of the Joint CDF

If you'd like, I can also create a **1–2 page "exam cheat sheet" version of Lecture 16** (formulas + typical exam steps for solving problems on joint PMF/PDF/CDF).

can u generate whole scribe in the canvas formate

Cse400 Lecture16 Scribe

CSE 400 – Fundamentals of Probability in Computing

Lecture 16 Scribe: Joint Random Variables and Joint Distributions

Based strictly on Lecture Slides by Dhaval Patel, PhD

---

1. Why Study Joint Random Variables?

In many real-world scenarios, we are interested in analyzing two or more random variables simultaneously instead of studying them independently. Joint analysis allows us to understand relationships, dependencies, and correlations between variables.

Examples of pairs of variables often studied jointly:

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate

- Number of Facebook friends and age of members

These examples motivate the need for **joint random variables**.

---

2. Applications Motivating Joint Random Variables

Example 1: Smart Transportation

Two random variables are defined:

X = Vehicle Speed
Y = Traffic Density

Question posed:

Can speed be high when the traffic density is also high?

Conclusion: The variables may be correlated.

Applications in Intelligent Transportation Systems (ITS):

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

Example 2: Wireless Network Performance

Two important random variables:

X = Signal-to-Noise Ratio (SNR)
Y = Data Throughput

Observation:

Throughput depends on SNR. Both values vary due to:

- Fading

- Interference
- Network congestion

Question:

If SNR is high, will throughput always be high?

Answer discussed: Not necessarily, because network congestion can affect throughput.

Example probability table:

SNR Level	High Throughput	Low Throughput
High	0.40	0.10
Medium	0.20	0.15
Low	0.05	0.10

Applications:

- 5G scheduling
- Adaptive modulation
- Network planning

Additional Examples

Example 3:

X = Rainfall
Y = Temperature

Application: Weather forecasting

Example 4:

X = Wind Speed
Y = Delivery Time

Application: Drone delivery and logistics systems

Example 5:

Rolling two dice:

X = Outcome of the first die
Y = Outcome of the second die

Example questions:

- X + Y = 7
- X > Y

These require the use of **joint probability distributions**.

---

3. Discrete Case – Joint PMF

For discrete random variables X and Y, probabilities are assigned to ordered pairs (x, y).

The **joint probability mass function (PMF)** is defined as:

p(x, y) = P(X = x, Y = y)

The joint PMF describes the probability that X takes value x and Y takes value y simultaneously.

---

4. Joint PMF Table Representation

A common way to represent a joint PMF is using a **table**.

Structure:

- Rows correspond to values of X
- Columns correspond to values of Y
- Each cell contains the probability p(x, y)

Properties of a joint PMF:

1. p(x, y) ⩾ 0 for all x and y

2. The sum of all probabilities equals 1

Sum over all x and y:

$\Sigma \Sigma p(x, y) = 1$

---

5. Example: Rolling Two Dice

Define random variables:

X = value of the first die
Y = value of the second die

Possible values:

$X \in \{1,2,3,4,5,6\}$

$Y \in \{1,2,3,4,5,6\}$

Total possible ordered outcomes:

36

Each pair (x, y) occurs with probability:

$P(X = x, Y = y) = 1/36$

Example events:

Event 1: $X + Y = 7$

Possible pairs:

(1,6) (2,5) (3,4) (4,3) (5,2) (6,1)

Number of favorable outcomes = 6

$P(X + Y = 7) = 6 / 36 = 1/6$

Event 2: $X > Y$

List of ordered pairs where X is greater than Y:

(2,1) (3,1) (3,2) (4,1) (4,2) (4,3) (5,1) (5,2) (5,3) (5,4) (6,1) (6,2) (6,3) (6,4) (6,5)

Total favorable outcomes = 15

$P(X > Y) = 15 / 36 = 5 / 12$

6. Continuous Case – Joint PDF

For continuous random variables X and Y, the distribution is described by a **joint probability density function (PDF)**.

Notation:

$f(x, y)$

Properties:

1. $f(x, y) \geqslant 0$
2. Total probability must equal 1:

$\int \int f(x, y)\, dx\, dy = 1$

Probability over a region R is computed as:

$P((X, Y) \in R) = \int \int_R f(x, y)\, dx\, dy$

This can be interpreted geometrically as the **volume under the surface f(x,y)** over the region R.

---

7. Worked Example (Continuous Joint PDF)

Given joint PDF:

$f(x, y) = 4xy$

for

$0 < x < 1 \quad 0 < y < 1$

This defines a joint distribution over the **unit square**.

Tasks:

1. Verify that f(x,y) is a valid joint PDF.

2. Visualize the event A = {X < 0.5, Y > 0.5}.
3. Compute P(A).

---

8. Example Solution

Step 1: Verify normalization

Compute the double integral over the support region.

$\int_0^1 \int_0^1 4xy \, dx \, dy$

First integrate with respect to x:

$\int_0^1 4xy \, dx$

$= 4y \int_0^1 x \, dx$

$= 4y \, [x^2 / 2]_0^1$

$= 4y \, (1/2)$

$= 2y$

Now integrate with respect to y:

$\int_0^1 2y \, dy$

$= [y^2]_0^1$

$= 1$

Thus the total probability equals 1, so the function is a valid joint PDF.

Step 2: Compute probability of event A

Event A:

X < 0.5 Y > 0.5

Probability:

P(A) = $\int_0$^{0.5} $\int$_{0.5}^{1} 4xy dy dx

First integrate with respect to y:

$\int_{0.5}^{1} 4xy\, dy$

$= 4x \int_{0.5}^{1} y\, dy$

$= 4x\, [y^2/2]_{0.5}^{1}$

$= 4x\, ((1 - 0.25)/2)$

$= 4x\, (0.75/2)$

$= 4x\, (0.375)$

$= 1.5x$

Now integrate with respect to x:

$\int_{0}^{0.5} 1.5x\, dx$

$= 1.5\, [x^2/2]_{0}^{0.5}$

$= 1.5\, (0.25 / 2)$

$= 1.5\, (0.125)$

$= 0.1875$

$= 3 / 16$

Therefore:

$P(A) = 3/16$

---

9. Joint Cumulative Distribution Function (Joint CDF)

Let X and Y be jointly distributed random variables.

The **joint cumulative distribution function** is defined as:

$F(x, y) = P(X < x, Y < y)$

This represents the probability that X is less than x and Y is less than y simultaneously.

---

10. Joint CDF – Continuous Case

If X and Y are continuous with joint density f(x,y), the joint CDF is obtained by integrating the joint PDF.

$F(x, y) = \int \int f(u, v)\, du\, dv$

The joint PDF can be recovered from the joint CDF using partial derivatives:

$f(x, y) = \partial^2 F(x, y) / (\partial x\ \partial y)$

---

11. Joint CDF – Discrete Case

If X and Y are discrete with joint PMF p(x, y), the joint CDF is computed by summing probabilities.

$F(x, y) = P(X \leqslant x, Y \leqslant y)$

This is obtained by summing:

$p(x_i, y_j)$

for all pairs satisfying:

$x_i \leqslant x$

$y_j \leqslant y$

---

12. Properties of the Joint CDF

1. The joint CDF is defined as:

$F(x, y) = P(X < x, Y < y)$

2. Range:

$0 \leqslant F(x, y) \leqslant 1$

3. The function is non-decreasing in both variables.
4. As x and y approach their maximum possible values, F(x,y) approaches 1.
5. As x and y approach their minimum possible values, F(x,y) approaches 0.

End of Lecture 16

Topics covered:

- Motivation for joint random variables
- Joint PMF for discrete random variables
- Table representation of joint distributions
- Example with rolling two dice
- Joint PDF for continuous random variables
- Worked example using density f(x,y)=4xy
- Joint cumulative distribution function

I've created the **full Lecture 16 scribe in Canvas format** so you can view and edit it easily.

You can now:

- Read it like a **structured document**
- Edit sections
- Ask me to **add diagrams, PMF tables, or formatting improvements**
- Ask me to convert it into **exam-ready condensed notes** or **LaTeX / PDF format**.